ALEX TANTHIPTHAM

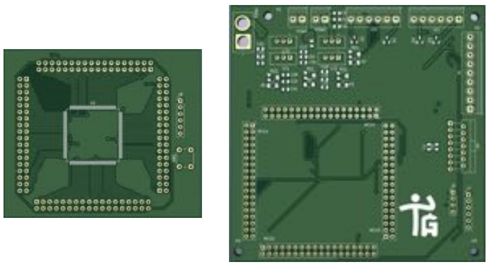
B.S. ELECTRICAL ENGINEERING
AT UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN

jirawatcharat@gmail.com
linkedin.com/in/alex-tanth/
(217) 931-1710

TAGALONG PERSONAL ASSISTANT ROBOT - UIUC

* Click here to learn more

Prototype MCU Breakout Board and Peripherals Board



Final Integrated MCU Circuit Board

Functional Prototype

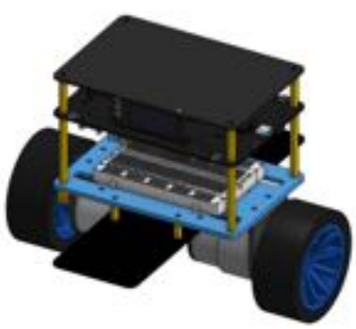
* Click to see the team!

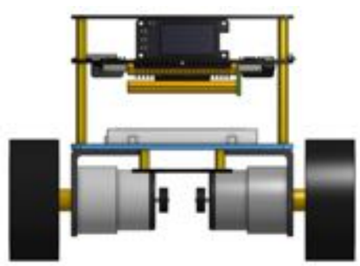
What?
- Design and assembled circuit components for cargo carrying robot capable of following users
- Aimed to aid elderly and differently abled in daily tasks

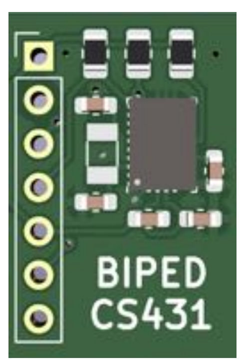
How?
- Designed and assembled modular MCU breakout PCB in **KiCAD** for fast and cost-efficient development
- Powered all subsystems from sealed lead acid batteries via linear and switching regulators

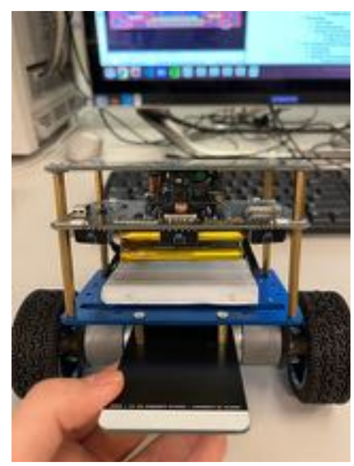
Results
- Awarded an **Honorable Mention** for Spring 2023 Senior Design Projects
- Approximately **16% the cost** of comparable market alternatives (Total retail cost of parts ~$180)

BIPED IMU REDESIGN - CPSIL @ UIUC









What?
- Redesigned dissatisfactory IMU breakout board, comparing different suppliers to ensure timely delivery
- Tested and verified units for hardware and software faults
- Reworked soldering, replaced faulty components

How?
- Identified issues during test runs
- Verified connectivity of components
- Designed IMU PCB for rapid development in **KiCAD**
- Assembled via hand and **reflow soldering**

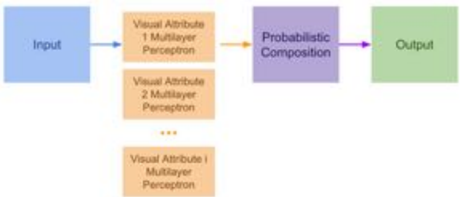
Results
- Increased number of classroom ready units for CS431 Embedded Systems from **3 to 20**
- Identified and resolved over **38** hardware and software faults
- Eliminated stand-still drift issues in units installed with new IMU board

ALEX TANTHIPTHAM

B.S. ELECTRICAL ENGINEERING
AT UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN

jirawatcharat@gmail.com
linkedin.com/in/alex-tanth/
(217) 931-1710

VISUAL ATTRIBUTES (VISAT) MODEL - CPSIL @ UIUC



Label: regulatory-right
Color: Blue
Shape: Circle
Symbol: Turn Right
Text: Undefined

GrabCut pre-processing for shape, color
Canny edge detection for symbol, text
Images composited for viewing

What?

- Engineer an image classification pipeline for interpretability and robustness against corruptions
- Implement dataset processing and internal tool scripts to improve performance and streamline development

How?

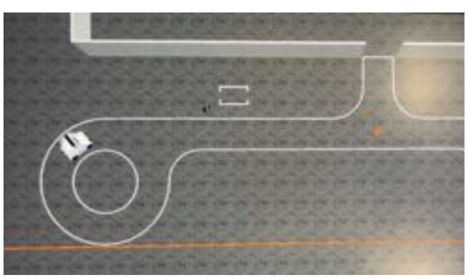
- Developed an ensemble of **PyTorch** MLP models for each attribute
- Designed 4 attribute labelling scheme
- Engineered pre-processing and attack scripts in **OpenCV** to improve accuracy and test for spurious correlations in attribute classification

Results

- Improved clean classification accuracy by up to **7.96%** over previous pipeline
- Reduced loss of accuracy under attacks by up to **17.83%**
- Reduce dataset processing runtime by up to **65%**

CLASSICAL CV LANE FOLLOWING - UIUC

Gazebo simulation of GEM EV and evaluation track at UIUC's Highbay facility



Camera output with detected lanes, their representative points and the plotted target path

Testing in vehicle under real world environmental and lighting conditions

What?

- Engineered a Python-based classical computer vision based **autonomous lane-following** system for a modified GEM e2 EV
- System generates GPS coordinates of path between lanes for use by a pure pursuit controller

How?

- Used **OpenCV** for lane segmentation and **Numpy** for path-planning,
- Used RGBD stereo camera for depth estimation and GNSS antenna output to generate coordinates relative to the current position
- Developed off site using **Gazebo** simulation and recorded ROS bags

Results

- Successfully completed the test track without the vehicle crossing any lanes within 2 minutes
- Achieved satisfactory real-world performance using 10% less reserved vehicle time than the average team